SOURAV PATTANAIK

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A fifth-year Integrated Master's student at the National Institute of Science Education and Research (NISER). My research focus is to combine wet-lab and dry-lab techniques for understanding complex biological phenomena with a multi-disciplinary approach and facilitate the translation of the research insights into innovative applications.

EDUCATION

2021 – 2026	**Integrated Master of Science in Biology (Minor degree in Chemistry)** National Institute of Science Education and Research (NISER), India **CGPA: 8.40 / 10.0**
2021	**12th Boards (CBSE)** | DAV School — **94.6%**
2019	**10th Boards (ICSE)** | Chinmaya Vidyalaya (E.M.) — **93.2%**

PUBLICATIONS

Rana S., **Pattanaik S.**, Thakur R., Joshi V., Jindal N., Tiwari R. R., & Sahoo G. C. (2026). Computational design and validation of a multi-epitope vaccine candidate against structural proteins of langya virus. Letters in Drug Design & Discovery, 100296. (https://doi.org/10.1016/j.lddd.2026.100296) **[Accepted]**

Pattanaik, S., Konkimalla, V. B. Directional Biological Network (DBN) Analysis Pro: automated target prediction analysis. **[Under Preparation]**

RESEARCH EXPERIENCE

2024 – 2026 — **MASTER'S THESIS** | Supervisor: Dr. V Badireenath Konkimalla, NISER, India

"Integrating network-based drug discovery and stimulus-responsive drug carrier for treating testicular cancer through an in silico approach."

- Designed a novel dual-stimulus responsive drug carrier using Intrinsically Disordered Proteins(IDPs).
- Developed a Random Forest Regression-based prediction model associated to IDPs with a predictive accuracy of 97.3%.
- Prepared a computational tool to upgrade static PPI.
- Screened for a potential drug molecule among the 17,94,286 molecules in the PubChem Database for binding to the therapeutic target.
- Techniques used: Structure Prediction, CG/Atomistic Molecular Dynamics Simulation, Protein-DNA Docking, Network analysis.

May – July 2025 — **SUMMER PROGRAMME** | Dr. Anbazhagan Kolandaswamy, RRMCH Bengaluru

"Discovery of Biomarkers for Sepsis Disease through comparative analysis of gene expression profiles."

- Optimized RNA extraction protocols and processed 52 patient blood samples, while maintaining high-integrity RNA extraction for quantitative profiling of 15 genetic targets(mRNA/miRNA), which were predicted to be differentially expressed using literature and Databases like miRTarBase.
- Identified and statistically validated 6 potential biomarkers(P-values < 0.05).
- Techniques learnt: qPCR, RT-PCR, Biostatistical analysis, Flow cytometry for immunophenotyping, FlowJo software for analysis, gel electrophoresis.

Jan – Mar 2025 — **REMOTE COLLABORATION** | Dr. Karoline Faust, KU Leuven

"Beta-testing of a new database associated with Systems Microbiology."

- Extracted, structured, and interpreted growth data of 10 different bacterial species.
- Contributed to Beta-testing of the new database.
- Acknowledged in research pre-print (doi:10.1101/2025.10.13.682118).

May – July 2024 — **SUMMER INTERNSHIP** | Dr. Sindhuprava Rana, ICMR-NIREH Bhopal

"Multi-Epitope Vaccine design against the structural proteins of Langya Virus."

- Used the protein sequence of 5 viral structural proteins to predict different properties like antigenicity, toxicity, etc.
- Suggested a novel strategy to include epitopes in the construct while maintaining solubility and biocompatibility.
- Published the research project as a co-first author (DOI mentioned above).

Nov – Dec 2023 **WINTER INTERNSHIP** | Dr. Abdur Rehman, Biochemistry lab, NISER
"Introduction to Life Cycle of Tetrahymena."
- Learned confocal microscopy and cell imaging using ImageJ.

May – July 2023 **SUMMER SCHOOL** | Dr. S Mirza, IIT Indore
"Principles of Drug Discovery and Development"
- Focus: Understanding the activity of anti-inflammatory drugs and stages of drug discovery.

May – July 2022 **SUMMER INTERNSHIP** | Dr. C S Purohit, Synthetic Chemistry Lab, NISER
"Multi-step Organic synthesis of 2,6-Bis{[bis(2-pyridinylmethyl)amino]methyl}-4-(1,1-dimethylethyl)phenol by aminomethylation."
- Techniques utilized: Synthesis, TLC, Column Chromatography, NMR.

CONFERENCES AND CONCLAVE

- **Conferences:**
  Quarterly Union for Academic Disclosure (QUAD)
  IAS-SRFP 36$^{\text{th}}$ Mid-Year meeting 2025
  InBix 2025.
- Participated in **the Springer Nature India Research Conclave** (2023)

TECHNICAL SKILLS

- **Computational:** Molecular Docking, MD Simulation, Sequence Alignment, Cell imaging, Machine Learning, Biological Network Interpretation, Pharmacophore Modeling, In-silico cloning
- **Wet-lab:** qPCR, RT-PCR, Gel electrophoresis, Plasmid/RNA/DNA isolation, Flow-cytometry, X-ray Crystallography, Protein purification, Cell culture, Confocal Microscopy, ELISA, Synthesis, NMR, Column Chromatography.
- **Additional:** LINUX, LaTeX, Java, Python, AI, Microsoft Office tools, GPU-based Computing, Websites, and Web tools development

AWARDS, FELLOWSHIPS AND SCHOLASTIC ACHIEVEMENTS

- **Competitive Exams Qualified:** National Entrance Screening Test 2021 [All India Rank (AIR) 263]; Graduate Aptitude Test in Engineering 2024 [AIR 1173]; Odisha Joint Entrance Examination 2021 [State Rank 3]; Odisha University of Agriculture and Technology 2021 [AIR 81]; National Eligibility cum Entrance Test 2021.
- **Scholarships/Fellowships:** Department of Atomic Energy (DAE)-DISHA Scholarship of 400000 INR(2021-2026); Indian Academy of Sciences-Summer Research Fellowship 30000 INR(May – July 2025).
- **Competitions:** Finalist of BENSPARK Ideathon, IISER Pune [2025]; Excelling ranks in Olympiads (English and Science) [2017-2019]; Won multiple inter-school and intra-college quizzes.

LEADERSHIP AND RESPONSIBILITY

- **Gym Treasurer (2024-2025):** Handled expenditure and bills amounting to almost 50000 INR for NISER Gym Club; Conducted Power-lifting events and served as a coordinator for multiple events.
- **Student Representative:** Member of the Disciplinary Action Committee (DAC) for Students (2025-2026). Contributed actively to policy making and adjudicated cases like substance abuse and harassment to ensure a safe campus environment.

HOBBIES AND EXTRA-CURRICULAR

- **Sports:** Represented NISER in Basketball (IISM 2023) and won inter-batch Basketball tournament (2024). Represented batch in Football (2024) and Badminton (2025) while winning bronze medals.
- **Cultural:** Won multiple dance and slogan writing competitions.
- **Other Interests:** Gym; Reading about novel therapeutic strategies in free time.

REFEREES

Dr. Sindhuprava Rana Scientist-C, ICMR-NIREH sindhup.rana@icmr.gov.in	**Dr. V. B. Konkimalla** Assoc. Professor, NISER badireenath@niser.ac.in	**Dr. A. Kolandaswamy** Assoc. Professor, RRMCH anbkol@rrmch.org